

MP DOCUMENTATION:

Technical Report:

“Pokerots: UP Tacloban Edition” is a 2D role-playing game (RPG) inspired by the mechanics and gameplay architecture of classic Pokémon emulator games. The project aims to recreate core gameplay systems such as character collection, turn-based combat, exploration, NPC interaction, and progression systems while integrating a modern thematic adaptation based on “Italian Brainrot” internet characters and the UP Tacloban Campus environment.

The game utilizes the UP Tacloban Campus as the primary explorable map, serving as the central game world where players navigate different locations, encounter enemy units, interact with non-playable characters (NPCs), and complete progression-based objectives. The map system is designed similarly to traditional tile-based RPG games, enabling structured movement, collision handling, event triggering, and environmental interaction through tile-based rendering techniques.

Instead of traditional Pokémon creatures, the game introduces custom AI-inspired fictional entities referred to as *“Pokerots.”* The implementation includes a turn-based battle system where players can capture, train, and strategically deploy Pokerots against opponents, and *“Gym Leader’s”* equivalents functioning as mini-boss encounters.

From a technical and software engineering perspective, the project is designed to demonstrate the practical application of core Object-Oriented Programming (OOP) concepts and fundamental game development principles using Java. The system incorporates encapsulation to securely manage game data and prevent unnecessary access to critical player and entity information. Inheritance and polymorphism are utilized to define reusable behaviors and specialized attributes among different entities, NPCs, and PokéRot types, enabling scalable and modular code architecture.

Additionally, multithreading techniques are implemented to simulate independent character behaviors and maintain smooth execution of concurrent game processes such as rendering, movement, and background updates. Arrays and multidimensional data structures are utilized for tile-based map generation, collision mapping, and exploration logic within the game environment.

The project also integrates File Input/Output (File I/O) operations to support persistent game storage, including saving and loading player progress, inventory states, and captured PokéRot data. A Graphical User Interface (GUI) is designed using Java Swing and related Java libraries to provide an interactive and immersive gameplay experience. Event handling mechanisms are implemented to process player inputs, movement controls, environmental interaction, and battle actions in real time.

Core systems developed in the project include:

- Tile-based map rendering and navigation
- Sprite and animation management
- Turn-based combat logic
- Entity and NPC interaction systems
- Inventory and capture mechanics
- Collision detection and movement handling
- Dialogue and event scripting systems
- Save/load functionality using File I/O
- GUI implementation using Java Swing
- Real-time game loop and rendering updates
- Event-driven input handling
- Multithreaded gameplay processes